

Shubhang Vagvala

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Summary

U.S. Citizen: Authorized to work anywhere in the U.S. without sponsorship.

Software engineer with hands-on experience shipping backend services, data pipelines, and developer tooling. Worked across order management, inventory pricing, distributed processing, search at scale, and AI-assisted developer workflows. Comfortable across concurrent API design, ETL and streaming pipelines, and CLI tooling, with working frontend exposure for full-stack feature delivery.

Work Experience

Stealth AI Startup

Founding AI Engineer

Nov 2025 – Apr 2026

San Francisco, CA

- Built a hybrid code indexing engine (**lexical + AST + semantic**) evolving into a **method-level tracing system** with cross-file **call graph traversal**, and exposed tracing and retrieval as **MCP-compatible tools** for **LLM agents**.
- Reduced **LLM context size** via **AST-based filtering** and **token-aware retrieval**, enabling **minimal-context**, **trace-backed reasoning**, and incorporated **git-diff-based impact analysis** for targeted review; benchmarked call graph resolution against **IntelliJ PSI**.

University of Massachusetts Amherst

Graduate Researcher

Sep 2024 – Aug 2025

Amherst, MA

- Studied **LLM robustness** to prompt perturbations in code generation; evaluated instruction-tuned models on **HumanEval** and **MBPP** across synonym, typo, and paraphrase augmentations, finding **pass@1 variance up to 23%**.
- Implemented a **work-stealing scheduler** for **Apache Flink** with load-scored redistribution to mitigate data skew; tested on **45M+ NYC taxi records**, improving runtime by **7.5%**, throughput by **8.4%**, and CPU efficiency by **11.8%**.
- Built and benchmarked a recommendation pipeline on **61M Amazon reviews**; trained collaborative filtering, content-based, **autoencoder**, and **two-tower** models with **PyTorch** and **FAISS** inference; two-tower outperformed baselines on sparse interactions.

MediBuddy

Software Engineer

Jun 2022 – May 2023

Bengaluru, India

- Migrated medicines bookings onto a unified **OMS** schema; built **SQL** migration scripts with dual-write cutover, state transition validation, and a cursor-paginated history API with reimbursement flows, achieving **zero data loss**.
- Architected an end-to-end vendor sync service, integrating **SFTP** ingestion and **CSV** adapters with exponential backoff retries; added a nightly reconciliation job with **Grafana** SLA alerts to surface refund gaps.
- Reduced order status API **p95 latency from ~2s to sub-300ms** by refactoring a 5-service fan-out into a concurrent **Java** workflow using **CompletableFuture**, per-service timeouts, and partial-response fallbacks.
- Optimized **Elasticsearch** substitute discovery with availability-weighted scoring and fuzzy matching; instrumented **React** search events for an **ML** pipeline with **~10% relevance improvement**.
- Engineered a geo-normalized address model using **Spring Boot** and **Google Maps APIs** with dual-path **Redis** caching and zipcode checks; improved first-attempt delivery success by **20%** and reduced support tickets by **17%**.

Swiggy

Software Engineer Intern

Jul 2021 – Jun 2022

Bengaluru, India

- Built a nightly **ETL** pipeline in **PySpark** on **Databricks** from a **CDC**-replicated **Hive** table to **Kafka**; reduced price-escalation incidents by **10%** on a **13M+ MAU** platform monitored with **Grafana**.
- Delivered a **gRPC** price lookup API in **Go** with dual-path lookups across availability and inventory services, using expiry-sorted batch matching and catalog fallback to reduce pricing errors.

Education

University of Massachusetts Amherst

M.S. in Computer Science, GPA: 3.97

Feb 2024 – May 2025

Coursework: Distributed Operating Systems, Software Engineering, Advanced Algorithms, Systems in Data Science

Birla Institute of Technology and Science Pilani

B.E. in Electronics and Communication Engineering, GPA: 3.61

Aug 2018 – May 2022

Coursework: Object-Oriented Programming, Data Structures and Algorithms, Microprocessors, Information Retrieval

Technical Skills

Programming Languages: Java, Python, Go, TypeScript, JavaScript, SQL, Bash.

Backend Development: Spring Boot, REST APIs, gRPC, GraphQL, Node.js, Express.js, FastAPI, Flask.

Frontend Development: React, Next.js, Redux, HTML/CSS, Swagger/OpenAPI, JWT.

Databases and Caching: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Elasticsearch, SQLite.

Data & ML: Apache Kafka, Apache Spark, Databricks, Apache Hive, PyTorch, FAISS, HuggingFace.

DevOps & Cloud: Docker, Kubernetes, Git, EC2, S3, Lambda, IAM, CloudWatch, CI/CD, Grafana.

AI / Developer Tooling: LLM APIs, MCP, Claude Code, Tree-sitter, SCIP, Zoekt, RAG, RAGAS.

Testing & Practices: JUnit, Mockito, Jest, Playwright, GitHub Actions, Agile/Scrum, TDD, SOLID.